5A - 6A - 10A - Atmosphere - EvolLand

Test 3 Master Test Review

41 Review Questions

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

MAGELLICA PRESS

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

REVIEW

Test 3 Master Test Review

41 questions worth focusing on as you prepare for Test 3, organized by subject. Each one tells you where to find the answer. Work through them in your own words, typing your answers on the lines.

The Earth's Atmosphere

1. What is the atmosphere? Excluding water vapor, name the four main gases that make it up, with their percentages.

Find answers in: Module 5A, 5AE1

2. Name the four layers of the atmosphere in order, from the ground up, and the "pause" at the top of each. What is a pause?

Find answers in: Module 5A, 5AE1

3. In which two layers does the temperature DECREASE with increasing altitude, and in which two does it INCREASE? What heats each of the two warming layers?

Find answers in: Module 5A, 5AE1

4. In which layer is the ozone layer, and what does it do? Which two gases in that layer protect life from deadly UV radiation? Why is ozone "good up high, bad nearby"?

Find answers in: Module 5A, 5AE1

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

5. What is the ionosphere, and where is it? Up to what frequency does it reflect radio waves?

Find answers in: Module 5A, 5AE1

6. What “sphere” harmlessly traps the charged particles of the solar wind?

Find answers in: Module 5A, 5AE1

7. What is Earth-based weather? Name the six main ingredients of Earth-based weather.

Find answers in: Module 5A, 5AE2

8. Name the four forms of precipitation, and describe each.

Find answers in: Module 5A, 5AE2

9. Are fog, dew, and clouds forms of precipitation? Explain.

Find answers in: Module 5A, 5AE2

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

10. In summer, what are cumulonimbus (storm) clouds primarily composed of?

Find answers in: Module 5A, 5AE2

11. What is a greenhouse gas? Name three. Why are nitrogen and oxygen NOT greenhouse gases?

Find answers in: Module 5A, 5AE2

12. What makes high and low pressure, and what weather does each bring?

Find answers in: Module 5A, 5AE2 (Homework #11)

Atmospheric Pressure Exercise

13. Define pressure and atmospheric pressure. How does atmospheric pressure change as you go higher, and why?

Find answers in: Atmospheric Pressure exercise

14. Give the value of one Atmosphere in millibars, psi, and inches of mercury (in Hg).

Find answers in: Atmospheric Pressure exercise

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

Ancient Atmospheres

15. Once early Earth cooled below the boiling point of water, what happened, for about how long, and what was the result?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

16. What was the Great Oxygenation Event, and which organisms caused it?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

EvolLand Exercise

17. About how many millions of years ago was the Cambrian Explosion, and what happened in it?

Find answers in: EvolLand exercise

18. As oxygen in the atmosphere rose, life moved from the sea to the land. About how high did oxygen peak, and when? What happened to flying insects?

Find answers in: EvolLand exercise (Module 5A, Homework #6)

19. In years, about how long ago did our species, Homo sapiens, evolve?

Find answers in: EvolLand exercise

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

The Nature of Light

20. What is optics?

Find answers in: Module 6A, 6AE1

21. What is the simple definition of light?

Find answers in: Module 6A, 6AE1

22. Light is described as a dual transverse, non-mechanical, self-propagating wave. What do “dual transverse,” “self-propagating,” and “non-mechanical” mean?

Find answers in: Module 6A, 6AE1

23. Rounded, what is the speed of light in a vacuum, in km/s and in miles/s?

Find answers in: Module 6A, 6AE1

24. Who first used astronomy to determine the speed of light, when, and how?

Find answers in: Module 6A

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

25. What does it mean to say laser light is coherent?

Find answers in: Module 6A, 6AE1

26. List the electromagnetic spectrum in order of increasing frequency. About what range of wavelengths can humans see? Are cosmic rays part of the spectrum?

Find answers in: Module 6A, 6AE1

27. What are the three ways light interacts with matter? Define each.

Find answers in: Module 6A, 6AE1

28. Name and define the five classical phenomena of light. Which model of light explains all five, and which one can the particle model adequately explain?

Find answers in: Module 6A, 6AE1

29. What is the difference between specular and diffuse reflection? Do all shiny objects exhibit specular reflection?

Find answers in: Module 6A

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

The Exploration of Space

30. What was Sputnik 1? Who launched it?

Find answers in: Module 10A

31. What was the first satellite the United States successfully launched? What did it discover?

Find answers in: Module 10A

32. What made Telstar 1 historic?

Find answers in: Module 10A

33. Who was the first person to orbit the Earth?

Find answers in: Module 10A

34. Who were the first and second Americans in space?

Find answers in: Module 10A

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

35. Who was the first American to orbit the Earth?

Find answers in: Module 10A

36. How many Apollo missions set out to land on the Moon, how many succeeded, and how many men have walked on the Moon?

Find answers in: Module 10A

37. What famous “first” in aviation belongs to Charles Lindbergh? In what plane, and from where to where?

Find answers in: Module 10A

38. Which uncrewed mission produced the first close-up images of Jupiter? What else was first about it?

Find answers in: Module 10A

39. Which uncrewed mission produced the first close-up images of Saturn?

Find answers in: Module 10A

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.

40. What were the Viking missions, and what was their key first?

Find answers in: Module 10A

41. Which spacecraft is the only one to have visited Uranus and Neptune?

Find answers in: Module 10A

Copyright © Magellica Press - R. Mahoney - 2026, All Rights Reserved.